Experiment No. 21

Write a Prolog Program to print particular bird can fly or not. Incorporate required queries.

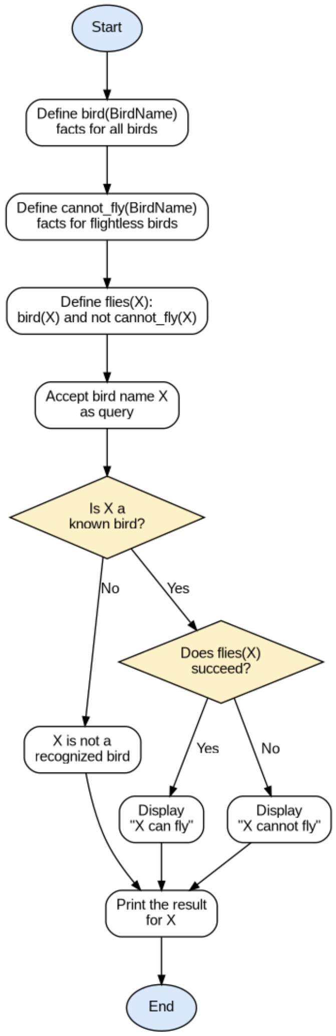
Aim

To write a Prolog program that determines whether a particular bird can fly or not, using facts and rules, and to incorporate the required queries to test the program for different birds.

Algorithm

1. Start.
2. Define facts bird(BirdName) for the set of known birds.
3. Define facts cannot_fly(BirdName) for birds that are unable to fly, such as penguin and ostrich.
4. Define the rule flies(X) which succeeds if X is a bird and X is not present in the cannot_fly facts.
5. Define the rule check_flight(X) that tests flies(X) for a given bird X.
6. If flies(X) succeeds, display the message "X can fly".
7. If X is a known bird but flies(X) fails, display the message "X cannot fly".
8. If X is not present in the bird facts at all, display the message "X is not a recognized bird".
9. Accept a bird name as a query to check_flight.
10. Repeat the check for multiple birds using separate queries.
11. Stop.

Flowchart



Program

```
bird(sparrow).
bird(eagle).
bird(pigeon).
bird(penguin).
bird(ostrich).
bird(crow).

cannot_fly(penguin).
cannot_fly(ostrich).

flies(X) :-
   bird(X),
   \+ cannot_fly(X).
```

```
check_flight(X) :-
    ( flies(X) ->
        format("~w can fly~n", [X])
    ; bird(X) ->
        format("~w cannot fly~n", [X])
    ; format("~w is not a recognized bird~n", [X])
    ).

main :-
    check_flight(sparrow),
    check_flight(penguin),
    check_flight(ostrich),
    check_flight(eagle),
    check_flight(cat).
```

Result

The program was executed successfully and the output was verified as correct.

```
?- main.
sparrow can fly
penguin cannot fly
ostrich cannot fly
eagle can fly
cat is not a recognized bird
true.
```

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal: